

Homogenous coordinate

Consider the effect of general 2 by 2 transformation applied to the origin

So for the origin, Origin is invariant under general 2 by 2 transformation.

This limitation is overcome by homogenous coordinates

It is necessary to be able to modify position of origin i.e. to transform every point in 2 dimension plane.

This can be accomplished by translating origin or any other point in 2 dimension plane

$$\begin{aligned}\text{If } x' &= ax + by + m \\ y' &= bx + dy + n\end{aligned}$$

In homogenous coordinate representation we add third coordinate to a point .

Instead of representing by a pair of number (x , y) each point is represented by a triple (x , y , h).

We say that 2 sets of homogenous coordinates (x , y , h) and (x' , y' , h) represent the same point if and only if one is multiple of another i.e. (2,3,6) , (4,6,12) are same points represented by different coordinate triples.

In order to transform a point (x , y) into homogenous representation we choose a non zero number 'h' and form a vector [hx , hy , h] and h is called scale factor or homogenous coordinate parameter.

For point [2,3] in 2 dimensional space it's representation in homogenous coordinated will be

$$\begin{aligned}[2, 3, 1] &\text{ for } h = 1 \\ [4, 6, 2] &\text{ for } h = 2 \\ [-2,-3,-1] &\text{ for } h = -1\end{aligned}$$

Homogeneous coordinates are widely used in computer graphics because they enable

Affine Transformation

Affine Transformation: a transformation that preserves collinearity (i.e. points lying on a line initially still lie on a line after transformation) and ratios of distances

It is a combination of single transformations such as translation, rotation or reflection on an axis

Hence the general transformation matrix is of the form

$$[T] = \begin{bmatrix} a & b & m \\ c & d & n \\ 0 & 0 & 1 \end{bmatrix}$$

hence for translation

$$P' = T(t_x, t_y) \cdot P$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}$$

now every point in 2 dimension plane every origin ($x = y = 0$) can be transformed.

Similarly, rotation transformation equation about coordinate origin EW

$$P' = R(\theta) \cdot P$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} x \cdot \cos\theta - y \cdot \sin\theta \\ x \cdot \sin\theta + y \cdot \cos\theta \\ 1 \end{bmatrix}$$

Scaling transformation relative to coordinate origin is

$$P' = S(s_x, s_y) \cdot P$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} x \cdot s_x \\ y \cdot s_y \\ 1 \end{bmatrix}$$